

Order matters

Production global routers process nets in an order

The idea

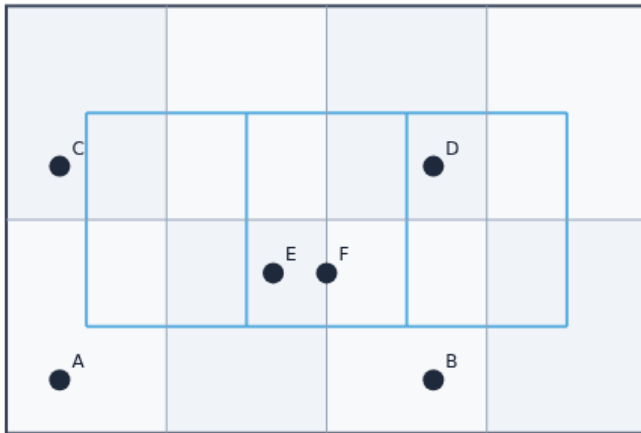
- Initialize empty usage
- Increment usage on every edge in the route
- Report final usage and edge_overflow summary

Net order

SEQUENTIAL GLOBAL

Net order

Step 1 / 5 · order · module04-01-sequential-global



Explanation

Route nets 0..5 in order; later nets see earlier usage.

- Six nets
- NETS array

Order matters

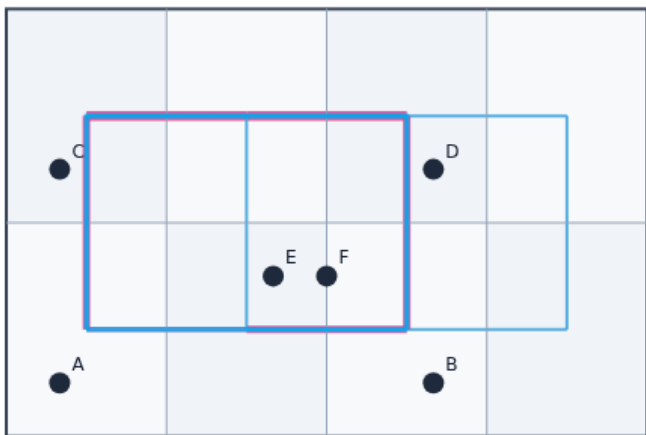
Net order

Pattern pass

SEQUENTIAL GLOBAL

Pattern pass

Step 2 / 5 · l-all · module04-01-sequential-global



Explanation

L-HV all nets documents baseline overflow.

- Fast pass
- Baseline

total≈2

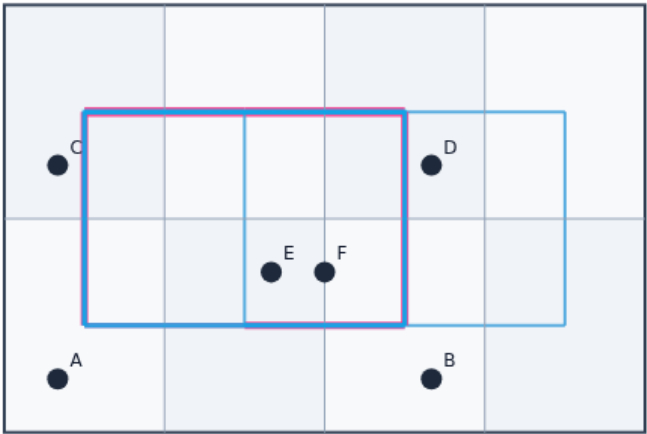
Pattern pass

Maze pass

SEQUENTIAL GLOBAL

Maze pass

Step 3 / 5 · maze-all · module04-01-sequential-global



Explanation

Sequential maze may redistribute usage.

- Penalties
- Compare totals

maze≈2

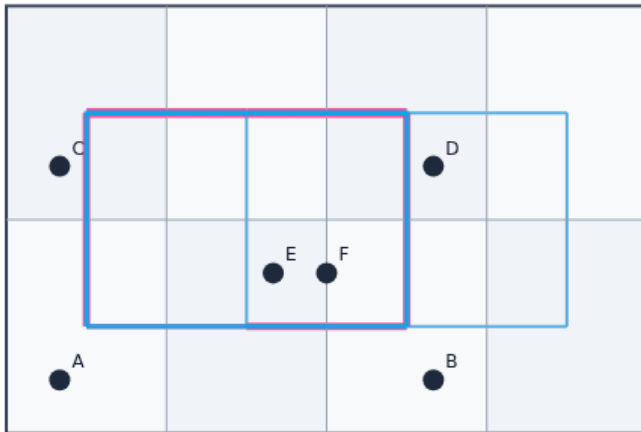
Maze pass

Clear overflow

SEQUENTIAL GLOBAL

Clear overflow

Step 4 / 5 · clear · module04-01-sequential-global



Explanation

Goal: total overflow 0 on the toy after moves or maze.

- Challenge lab
- HPWL optional

Learner state

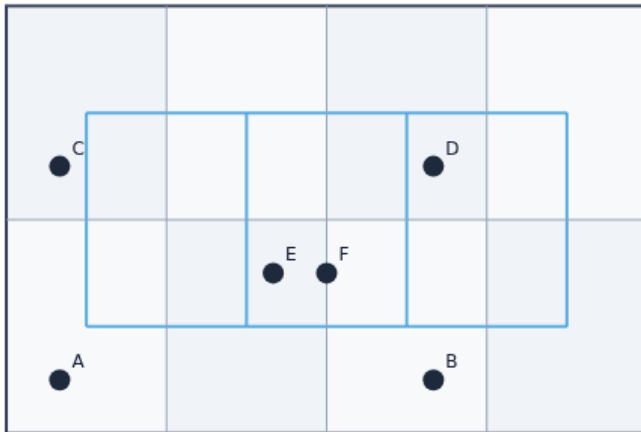
Clear overflow

Handoff

SEQUENTIAL GLOBAL

Handoff

Step 5 / 5 · done · module04-01-sequential-global



Explanation

Global routing feeds detailed routing and DRC-clean paths.

- Edges + caps
- Course wrap

Next: `learn_routing`

Handoff

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Sequential global routing

INTERACTIVE TOOL

Sequential global routing

Route all six nets in order; clear overflow thresholds on the tiny instance.

Your job: route all 6 nets. Try L-HV, L-VH, maze until overflow clears.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntroSix nets

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

StartShow hintCheckNext challengeStop checkingIdle

A B C D E F←→↑↓Reset to starterReveal golden (study)Route L-HV

Route L-VHRoute mazeRip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

Implement track

- Implement ``route_nets`` and ``route_nets_with_routes``
- Print usage and overflow after routing all six nets
- Compare with maze mode on total overflow

Pitfalls

- Parallel deposit without order, hides rip-up motivation
- Ignoring multi-pin net in order list
- Resetting usage between nets accidentally

Your turn

- Complete sequential global routing
- Offline compare and wrap come next